

Programme	BSc Mathematics Honours			
Course Title	INTRODUCTION TO PYTHON AND SCIENTIFIC COMPUTING			
Type of Course	SEC – Double Major			
Semester	IV			
Academic Level	200-299			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	3	3	-	45
Pre-requisites	(1) Basic knowledge to start a desktop/laptop computer. (2) A basic course in calculus with an understanding of differential and integral calculus. (3) A basic course in matrix algebra (higher secondary level)			
Course Summary	This course introduces the fundamentals of Python with a focus towards mathematical programming. Getting started with Python, Various Interfaces, Variables, Modules, Loops, Lists, Tuples, Functions, Branching, Input and Output, Arrays and Plotting, Dictionaries and Strings and finally Classes and Object-Oriented Programming are introduced. Using the Python programming structure, an introduction to the advanced mathematics software SageMath is given in the last part of the course. Various practical problems making use of concepts from calculus and linear algebra are to be solved using the SageMath software in the open-ended practical part so that the students will come to know how to apply software to answer and compute typical problems from these subjects.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand Basics of Python Programming.	U	C	Internal Exam/ Assignment/ Practical Assessment / Viva/ End Sem Exam
CO2	Intermediate Level Concepts such as Object-Oriented Programming.	An	P	Internal Exam/ Assignment/ Practical Assessment / Viva/ End Sem Exam
CO3	Scientific Computation using SageMath.	E	P	Internal Exam/ Assignment/ Practical Assessment / Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	1. Introduction to Scientific Programming with Python, Joakim Sundnes, Simula SpringerBriefs on Computing, 2020, ISBN: 978-3-030-50356-7. Open Access: https://link.springer.com/book/10.1007/978-3-030-50356-7 2. Sage for Undergraduates, 2nd Ed., Gregory V. Bard, 2022, American Mathematical Society, 2022. ISBN: 978-1470411114. 2014 Online Ed: http://www.people.vcu.edu/~clarson/bard-sage-for-undergraduates-2014.pdf			
Module	Unit	Content	Hrs (36+ 9)	Marks Ext: 50
I	Python Basics (Text 1, Ch. 1, 2, 3, 4.)		8	Min.10
	1	Getting Started (Ch 1). Programming Simple Mathematics (Sec 2.1). Variables and Variable Types (Sec 2.2).		
	2	Formatting Text Output. Importing Modules. (Sec 2.3, 2.4).		
	3	Loops and Lists. Loops for Automating Repeated Tasks. Using Lists to Store Sequences of Data. (Sec 3.1, 3.2, 3.3).		
	4	Iterating over a List with a for Loop.. Nested Lists and List Slicing. (Sec 3.4, 3.5).		
	5	Tuples. (Sec 3.6)		
II		Functions, Branching, I/O, Modules.	8	Min 10
	6	Programming with Functions.. Function Arguments and Local Variables. Default Arguments and Doc Strings. (Sec 4.1, 4.2, 4.3)		
	7	If Tests for Branching the Program Flow. Functions as arguments to Functions. (Sec 4.4, 4.5)		
	8	Solving Equations with Python Functions. (Sec 4.6)		
	9	Writing Test Functions to Verify Programs (Sec 4.7).		
	10	User Input and Error Handling. Reading Input User Data. Reading Data from Files. Writing Data to Files. (Sections 5.1, 5.3, 5.4. Section 5.2 omitted).		
	11	Handling Errors in Programs. (Sec 5.5)		
	12	Making Modules. (Sec 5.6)		

III	More Data Structures, Plotting (Text 1, Ch. 6, 7).		7	Min 10
	13	Arrays and Plotting. Numpy and Array Computing. Plotting Curves with Matplotlib. (Sec 6.1, 6.2)		
	14	Plotting Discontinuous and Piecewise Defined Functions. (Sec 6.3).		
	15	Dictionaries and Strings. Examples: A Dictionary for Polynomials, Reading File Data to a Dictionary. (Sec 7.1 7.2, 7.3),		
	16	String Manipulation (Sec 7.4).		
IV	Classes and Object-Oriented Programming. (Text 1, Ch. 9, 10.)		7	Min 10
	17	Basics of Classes. (Sec 8.1)		
	18	Protected Class Attributes, Special Methods. Example: Automatic Differentiation of Functions. (Sec 8.2, 8.3, 8.4).		
	19	Test Functions for Classes. Example: A Polynomial Class. (Sec 8.5, 8.6).		
	20	Class Hierarchies and Inheritance. Example: Classes for Numerical Differentiation, Integration. (Sec 9.1, 9.2, 9.3).		
V	Open-Ended Lecturer’s selections of 15 sessions of 2 hours each from below. Miscellaneous Python Exercises 1. Pitfalls of Programming, Text 1, Section 2.5. 2. Familiarize various Python runtime environments and IDEs like IDLE, Spyder, VS Code, Virtual Environments, Jupyter Notebook, Google Colab, Anaconda/Miniconda/Mamba, Replit. 3. Familiarize various documentation websites and how to refer to the syntax and implementation of a Python concept or Package. 4. Case studies from Reference 2:, Income Tax Calculator (page 38), Investment Report (p. 73), Approximating Square Roots. (p. 92), Text Analysis (p. 126), Generating Sentences (p. 150). Sagemath			

1. Getting and installing sagemath in Windows, Ubuntu OS Using sagemath using cocalc (online).
2. Using Sage as a Calculator, Using Sage with Common Functions, Using Sage for Trigonometry (Text 2, sections 1.1, 1.2, 1.3).
3. Using Sage to Manipulate Polynomials (Text 2, section 1.7)
4. Matrices and Sage-A First Taste of Matrices, Doing the RREF in Sage (Text 2, section 1.5)
5. Using Sage for 2-D graphs (Text 2, section 1.4)
6. The Derivative, Slope of Tangent, Higher-Order Derivatives (Text 2, section 1.11))
7. Antiderivatives (Indefinite Integral), Definite Integrals, Improper Integrals (Text 2, sec 1.12, upto sec 1.12.6)

Sympy (Reference 3).

1. Sympy Introductory Tutorial.
2. Solve an equation algebraically.
3. Solve a system of equations algebraically.
4. Solve one or a system of equations numerically.
5. Find the roots of a polynomial symbolically or numerically.
6. Solve a matrix equation algebraically.
7. Solve a Diophantine equation algebraically.
8. Solve an ODE algebraically.

More Numpy and Data Visualization (Reference 1: Chapter 3, 4)

1. Numpy Functions: arange, linspace, zeros, ones, random.random, reshaping. (Sec 3.1.1 to 3.1.6). Copying, Saving and Restoring, Slicing, Arithmetic Operations. (Sec 3.1.7 to 3.1.10).
2. Matplotlib Module: 2D Plots, Polar Plots, Pie Charts, Multiple Plots. (Sec 4.1)
3. Sine function and friends, Circle, Parametric Plots, Error Bars. (Sec 4.2)
4. Simple 2D Animation (Reference 1, Section 4.4), Making a movie of a Plot (Text 1, Section 4.4)
5. Famous Curves: Astroids, Ellipse, Spirals of Archimedes and Fermat (Reference 1, Sec 4.5)
6. 2D Plots and Fractals (Reference 1, Section 4.6)
7. 3D Plots (Reference 1, Section 4.7)

Numerical methods using SageMath (Reference 5: Chapter 7)(7.1 - 7.10, 7.12)

- 1) Evaluate a Taylor series numerically.
- 2) Interpolate a function using
 - a) Newton's forward interpolation.
 - b) Newton's backward interpolation.
 - c) Lagrange's Interpolation.

- d) Newton's General Interpolation.
- 3) Find integral of function using
 - a. Trapezoidal Rule
 - b. Simpson's 1/3-rule
- 4) Find derivative of function numerically.
- 5) Solve first order differential equations numerically.
 - a) Euler method
 - b) Fourth order Runge-Kutta method
- 6) Solve algebraic equations numerically.
 - a) The Bisection method
 - b) Regula Falsi Method

References

1. Python for Education, Ajith Kumar B. P., 2023
<https://scischool.in/python/pythonForEducation.pdf>
2. Fundamentals of Python First Programs, Kenneth A Lambert, 2 Ed., Cengage, 2018.
3. SymPy Tutorial: <https://docs.sympy.org/latest/tutorials/intro-tutorial/index.html>
Solving Equations: <https://docs.sympy.org/latest/guides/solving/index.html>
4. Computational Mathematics with SageMath, Paul Zimmermann, Alexandre Casamayou,
<https://www.sagemath.org/sagebook/english.html>
5. SageMath Advice For Calculus, Tuan A. Le and Hieu D. Nguyen,
<https://users.rowan.edu/~nguyen/sage/SageMathAdviceforCalculus.pdf>
6. Sagemath Reference: <https://doc.sagemath.org/>

Programming Resources

1. Python official website: <https://www.python.org>
Documentation: <https://docs.python.org/>
2. Spyder official website and documentation, <https://www.spyder-ide.org/>
3. MIT Courseware, Getting Started: Python and IDLE,
<https://web.mit.edu/6.s189/www/handouts/GettingStarted.html>
4. Jupyter Notebook, <https://jupyter.org/>
5. Google Colaboratory (colab), <https://colab.google/>
6. Visual Studio Code: <https://code.visualstudio.com>,
Documentation: <https://code.visualstudio.com/docs>
VS Code for Web: <https://vscode.dev/>
7. Replit, <https://replit.com/>
8. Python Virtual Environments: <https://docs.python.org/3/tutorial/venv.html>
9. Anaconda, Miniconda and Mamba.
Anaconda: <https://docs.anaconda.com/free/anaconda/>

Miniconda: <https://docs.anaconda.com/free/miniconda>

Mamba: <https://mamba.readthedocs.io/en/latest/>

10. SageMathCloud at Cocalc: <https://cocalc.com>

Documentation: <https://doc.cocalc.com/>

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	3	2	1	3	2	3	3	2	1	2
CO 2	3	3	2	2	3	2	3	3	2	1	2
CO 3	3	3	3	3	3	1	3	3	3	1	3

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High